

Question	Answer	Mark	Mark scheme	Additional guidance																																																
19	60 and reason	B1 M1 M1 A1 C1	for 432.5 or 437.5 or 9.75 or 9.85 or 2.45 or 2.55 for a correct process to find a bound for T eg [LB of w] $\div$ [UB of $a - \text{LB of } c$] where 432.5 ,, [LB of w] $<$ 435 and 9.8 $<$ [UB of a] ,, 9.85 and 2.45 ,, [LB of c] $<$ 2.5 or [UB of w] $\div$ [LB of $a - \text{UB of } c$] where 435 $<$ [UB of w] ,, 437.5 and 9.75 ,, [LB of a] $<$ 9.8 and 2.5 $<$ [UB of c] ,, 2.55 for a correct process to find both LB and UB bound for T eg [LB of w] $\div$ [UB of $a - \text{LB of } c$] where 432.5 ,, [LB of w] $<$ 435 and 9.8 $<$ [UB of a] ,, 9.85 and 2.45 ,, [LB of c] $<$ 2.5 and [UB of w] $\div$ [LB of $a - \text{UB of } c$] where 435 $<$ [UB of w] ,, 437.5 and 9.75 ,, [LB of a] $<$ 9.8 and 2.5 $<$ [UB of c] ,, 2.55 (dep on all previous marks) for 58.44(5...) and 60.76(3...) with both values clearly coming from working with correct values for 60 from 58.44... and 60.76... and statement that both LB and UB round to 60	<table><tr><th>Lette r</th><th>Given</th><th>LB</th><th>UB</th></tr><tr><td>w</td><td>435</td><td>432.5</td><td>437.5</td></tr><tr><td>a</td><td>9.8</td><td>9.75</td><td>9.85</td></tr><tr><td>c</td><td>2.5</td><td>2.45</td><td>2.55</td></tr></table> UB <table><tr><th>Lette r</th><th>Given</th><th>LB</th><th>UB</th></tr><tr><td>w</td><td>435</td><td>432.5</td><td>437.5</td></tr><tr><td>a</td><td>9.8</td><td>9.75</td><td>9.85</td></tr><tr><td>c</td><td>2.5</td><td>2.45</td><td>2.55</td></tr></table> LB <table><tr><th>Lette r</th><th>Given</th><th>LB</th><th>UB</th></tr><tr><td>w</td><td>435</td><td>432.5</td><td>437.5</td></tr><tr><td>a</td><td>9.8</td><td>9.75</td><td>9.85</td></tr><tr><td>c</td><td>2.5</td><td>2.45</td><td>2.55</td></tr></table> Accept bounds rounded or truncated to at least 4 sf	Lette r	Given	LB	UB	w	435	432.5	437.5	a	9.8	9.75	9.85	c	2.5	2.45	2.55	Lette r	Given	LB	UB	w	435	432.5	437.5	a	9.8	9.75	9.85	c	2.5	2.45	2.55	Lette r	Given	LB	UB	w	435	432.5	437.5	a	9.8	9.75	9.85	c	2.5	2.45	2.55
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